

Associator Spectroscopy — OSF-Ready Preregistration

Executive Summary

We propose an experiment to **detect and repair closure defects** in coarse-grained models by treating **non-associativity** of the macro merge operator as a measurable signal. Using droplet **coalescence** as the canonical testbed, we run a two-path “associator spectroscopy” protocol— $(AB)C$ vs. $A(BC)$ —and quantify gaps at the level of means and distributions. When gaps persist for a macro variable (e.g., N , droplet count), we fit the **minimal augmentation** C that restores associativity for $\Psi = (N, C)$. We pre-register **success/failure criteria**, a **leak-free model-selection pipeline**, and an **information-bottleneck objective** that operationalizes minimality.

Hypothesis

For coalescence-like systems, the macro variable N (droplet count) is **not closed**: the induced binary operation $x \oplus y$ on $x = N$ is **non-associative**. There exists a **low-dimensional** augmentation C such that $\Psi = (N, C)$ yields an **approximately associative** $\oplus$ within pre-registered tolerances.

System

- Emulsion droplets generated on chip; merges via Brownian contact.
 - Optional surfactant to modulate interaction strength (suppression factor $\alpha \in [0, 1]$).
 - Time-windowed stages implemented in **identical, isolated microchannels** to avoid environmental carryover between the left/right parenthesizations.
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Macro Variables

- **Primary:** N (droplet count).
 - **Candidate augmentation C (1–6 scalars):** second/third moments ($M_2 = \sum m_i^2$, M_3), polydispersity index (PDI), and quantiles ($q_{.25}$, $q_{.5}$, $q_{.75}$). Total mass $M_1 = \sum m_i$ is monitored (conserved) and used for the physical null.
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Triplet “Associator Spectroscopy” Protocol

1. **Triplets:** Prepare matched samples (A, B, C) with equal total mass and closely matched size distributions.

2. **Paths:** Run two parenthesizations for equal time windows:
3. Left: $(AB)C$ = merge A with B , then with C .
4. Right: $A(BC)$ = merge B with C , then with A .
5. **Outcomes:**
6. Mean gap: $A_E = \mathbb{E}[X]_{(AB)C} - \mathbb{E}[X]_{A(BC)}$, for $X \in \{N, \Psi\}$.
7. Distributional gap: $A_D = W_1(\nu_{(AB)C}, \nu_{A(BC)})$ (Wasserstein-1).
8. **Controls:** 4 heterogeneity levels (mono→poly) × 3 surfactant levels.

Matching Tolerance

Require **Wasserstein-1** distance between droplet-radius distributions to be small:

- Accept triplets only if $W_1(r_A, r_B) < \tau$ and $W_1(r_A, r_C) < \tau$.
- Pilot targets: median-radius mismatch < **2%** and $W_1 < 0.02 \times$ median radius.

To declare a closure defect for the macro variable $N**$, we require that both $|A_E(N)| > 3\sigma_{\text{null}}$ and $A_D(N) > 3\tilde{\sigma}_{\text{null}}$ in at least 3 of the 4 heterogeneity levels, after familywise error correction (Holm-Bonferroni, $**\alpha = 0.01**$) across all 4 levels.**

Definitions:

- σ_{null} ********: standard deviation of $A_E(M_1)$ across replicates under the **physical null** (mass conservation).
- $\tilde{\sigma}_{\text{null}}$ ********: standard deviation of A_D from the **synthetic associative pipeline** (full histogram as state; exact concatenation; same measurement/analysis stack).

Declare closure repaired if **both** $\mathbb{E}|A_E(\Psi)| \leq \epsilon_E$ and $\mathbb{E}A_D(\Psi) \leq \epsilon_D$ with $\epsilon_E = 2\sigma_{\text{null}}$ and $\epsilon_D = 2\tilde{\sigma}_{\text{null}}$ **across all conditions.**

Power & Replicates

- Pilot determines $\sigma_{\text{null}}, \tilde{\sigma}_{\text{null}}$ and standardized effect size d .
- With $d \approx 0.5$, two-sided $\alpha = 0.01$, power 0.8 → **n = 50-70 triplet replicates per condition** (heterogeneity × surfactant). ****Total $\approx **4 \times 3 \times n$ triplets.**

Stage 1 — Associator Measurement. Collect all triplets; split 60/20/20 into **Discovery / Validation / Holdout.**

Stage 2 — Minimal Augmentation Search. For each candidate C :

- Fit **macro law** G on the *first-merge* outcomes in Discovery. Model class: quadratic polynomials in Ψ ; escalate only if needed to a small ReLU MLP (≤ 64 hidden units).
- Evaluate the pair (C, G) on **Validation** by in-silico rolling the triplet protocol and measuring $A_E(\Psi), A_D(\Psi)$.
- Select the **first** (C, G) meeting tolerances, prioritizing the **Information Bottleneck** objective (below).

Stage 3 — Final Reporting & Robustness. Freeze (C, G) . Report triplet and **pentagon** (four-merge) residuals on **Holdout**; perform **ablations** (drop each component of C).

Information Bottleneck Minimality. Primary objective:

$$\min_{C, G} I(\Sigma; C) \quad \text{s.t.} \quad \mathbb{E}[A_E(\Psi)] \leq \epsilon_E, \mathbb{E}[A_D(\Psi)] \leq \epsilon_D.$$

- **Implementation note:** Use a small MLP encoder with **spectral normalization** and early stopping; estimate $I(\Sigma; C)$ from the **available microstate features** (full measured histogram). Secondary minimality criteria: smallest $|C|$; predictive sufficiency (next-step macro $\perp$ micro $|\Psi$).

Extended Checks

- **Pentagon identity:** four-step merges; report the “pentagon residual” as a higher-order associativity test.
- **Cocycle residual:** if $x \oplus y = x + y + I(x, y)$, report $\Delta = [I(x, y) + I(x \oplus y, z)] - [I(y, z) + I(x, y \oplus z)]$ before/after augmentation.

Failure Modes (to Report Regardless)

- No low-dimensional C achieves tolerance $\rightarrow$ evidence of **intrinsic non-closability** at usable scales (strong-emergence claim).
- Excess path sensitivity only under extreme heterogeneity $\rightarrow$ delineates **domain of validity** for the macro model.

In-Silico Pilot (for Calibration)

Simulator. Time-windowed Gillespie-style coalescence with a Brownian-like kernel; matched triplet generation using a radius-based W_1 tolerance; equal time windows per stage.

Quick readout (n=8 per cell; small, illustrative):

polydispersity $\sigma_{\log}$	surfactant α	$A_E(N)$	$A_D(N)$
0.15	1.0	-0.125	0.071
0.15	0.5	-0.125	0.357
0.45	1.0	0.250	0.214
0.45	0.5	0.250	0.214

Mass-null spread was effectively zero at this scale ($\sigma_{\text{null}} \approx 5 \times 10^{-16}$); matching quality $\overline{W}_1$ on radii ≈ 0.002 – 0.007 . **Directionality matches expectations:** greater heterogeneity $\rightarrow$ larger associator signal; surfactant suppression modulates the gap.

Use this pilot to set τ , n , and ϵ realistically before wet-lab work; then rerun power with stabilized null spreads.

OSF Checklist

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External References (selected)

- **Coalescence / Smoluchowski.**

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- **Microfluidic droplets.**

- Teh, S.-Y., Lin, R., Hung, L.-H., & Lee, A. P. (2008). *Droplet microfluidics*. **Lab on a Chip**, **8**(2), 198–220.

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Where possible, references above include editions or sources with open PDFs/DOIs for long-term OSF linking.

Appendix: Notation Cheatsheet

- $\Phi(\sigma)$: chosen macro map (here, N). $\Psi(\sigma) = (\Phi(\sigma), C(\sigma))$.
- $\oplus$: induced macro merge under Ψ via fitted law G .
- A_E : mean associator gap; A_D : distributional associator gap (Wasserstein-1).
- $\sigma_{\text{null}}, \tilde{\sigma}_{\text{null}}$: null spreads (physical & synthetic).
- ϵ_E, ϵ_D : closure tolerances; **IB**: information bottleneck objective.

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